For the Curious: '/var/run/docker.sock'



The /var/run/docker.sock file is a socket created by the Docker daemon to allow other processes to communicate with it via a (mostly) RESTful API.[8] We can use curl to try it out—for example:

```
$ SOCKET=/var/run/docker.sock
$ ENDPOINT=http:/v1.37/containers/json
$ docker-machine ssh do-manager-1 \
     "curl --unix-socket $SOCKET $ENDPOINT"
```

You should see JSON output listing the various containers.

See the API documentation[9] and version history[10] for further details.

Next come lines 7–9. We haven't seen the deploy attribute for a service before.[11] Swarm uses attributes under this to specify various deployment-related configuration options. Here we specify what's known as a *placement constraint*:[12]

```
[node.role == manager]
```

This says, "Only deploy this service to nodes acting as a swarm manager." This is necessary because only a swarm manager will have the full information about the swarm that the visualizer needs to display.

OK, let's deploy the new visualizer service to our swarm:

```
$ docker stack deploy -c docker-stack.yml myapp
Updating service myapp_database (id: tjlxqvm3vxop9flor3u4q21y2)
Creating service myapp_visualizer
Updating service myapp_db-migrator (id: zy14mg5nosigzi9i0ysr2f51y)
Updating service myapp_web (id: vo89apdhl4y0sfobb47ly8ctp)
Updating service myapp_redis (id: sxum5csqxwcujgy0g7418fv9y)
```

Wait until the visualizer is running—the following command will show replicas "1/1" for the visualizer service when it is:

```
$ docker stack services myapp
ID   NAME               MODE  REP…  IMAGE                 PORTS
sx…  myapp_redis        rep…  1/1   redis:latest
tj…  myapp_database     rep…  1/1   postgres:latest
vo…  myapp_web          rep…  1/1   robisenberg/myapp_w…  *:80->3000/tcp
za…  myapp_visualizer   rep…  0/1   dockersamples/visua…  *:8080->8080/…
zy…  myapp_db-migrator  rep…  0/1   robisenberg/myapp_w…
```

8. https://docs.docker.com/develop/sdk/examples/
9. https://docs.docker.com/engine/api/v1.37/
10. https://docs.docker.com/engine/api/version-history/
11. https://docs.docker.com/compose/compose-file/#deploy
12. https://docs.docker.com/engine/reference/commandline/service_create/#specify-service-constraints---constraint